STA 209: Introduction to Time Series Analysis

Homework 4

Rento Saijo

April 1, 2026

Table of Contents

Disclosure

GPT-5.4 Codex was used to create the `YAML` portion and some `LaTeX` code to format the text/equations nicely. Page formatting code was also provided by Derin Gezgin. In the setup chunk, chunk options were set for the document. See the original `RMD` file here for more details.

Problem 1

Problem 1

Consider the datasets `sales` and `lead` in package `astsa`.

Cell 1

```
# Load data.
sales <- astsa::sales
lead  <- astsa::lead
```

The package help states that both series contain 150 observations and are taken from Box and Jenkins (1970). The `sales` series is monthly sales data and the `lead` series is the corresponding leading indicator.

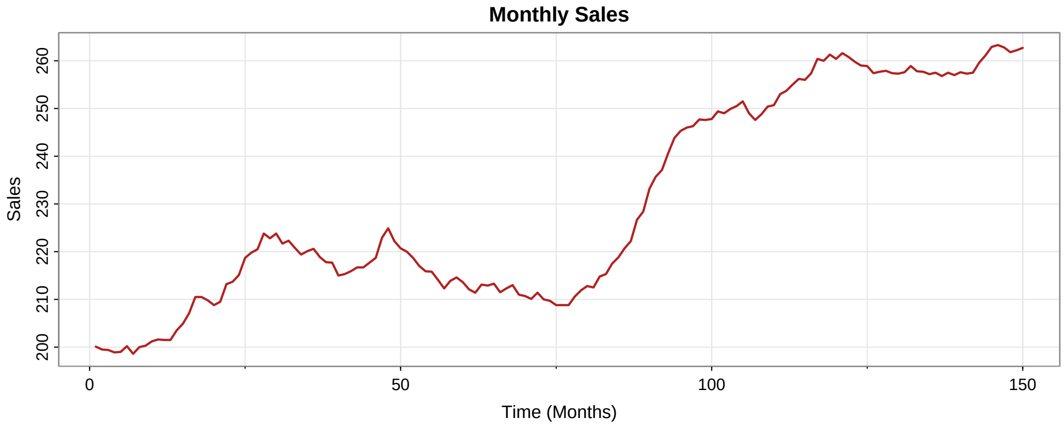
Problem 1 (a)

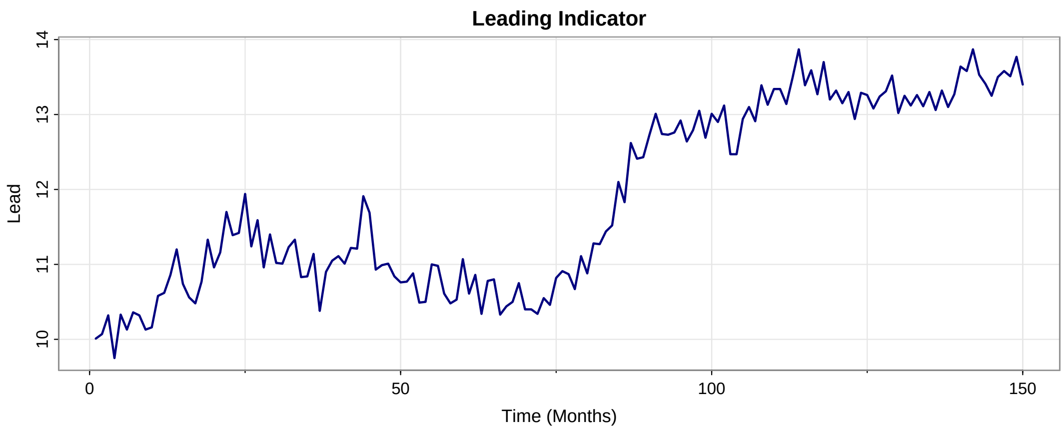
Problem 1 (a)

Plot and comment on the main features of these two time series datasets.

Cell 2

```r
# Plot the two original series in separate panels.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 1))
astsa::tsplot(
  sales,
  main = 'Monthly Sales',
  xlab = 'Time (Months)',
  ylab = 'Sales',
  col = 'firebrick',
  lwd = 2
)
astsa::tsplot(
  lead,
  main = 'Leading Indicator',
  xlab = 'Time (Months)',
  ylab = 'Lead',
  col = 'navy',
  lwd = 2
)
```





The `sales` data is a monthly sales time series with 150 observations, so it covers 150 months, or about 12.5 years. The `lead` data is the monthly leading-indicator time series for the same 150 months. The mean behavior of both series is time-dependent and shows an overall upward trend. For `sales`, the trend is positive and somewhat nonlinear because the series rises overall, pauses or dips at several points, then increases sharply around the middle-to-late part of the sample before flattening somewhat near the end. For `lead`, the upward movement is also clear, but it evolves on a smaller scale and with more month-to-month irregular fluctuation than `sales`. There is no clear repeated seasonal pattern in either plot despite the monthly observation frequency. The variability is moderate in both series, with statistical range $\max(\texttt{sales}) - \min(\texttt{sales}) = 64.7$ and $\max(\texttt{lead}) - \min(\texttt{lead}) = 4.12$. For both, the spread appears fairly stable over time, so there is no strong visual evidence of fanning in or out. The `sales` series does not look especially volatile, while `lead` shows more short-run fluctuation, but still no strong volatility clustering.

Problem 1 (b)

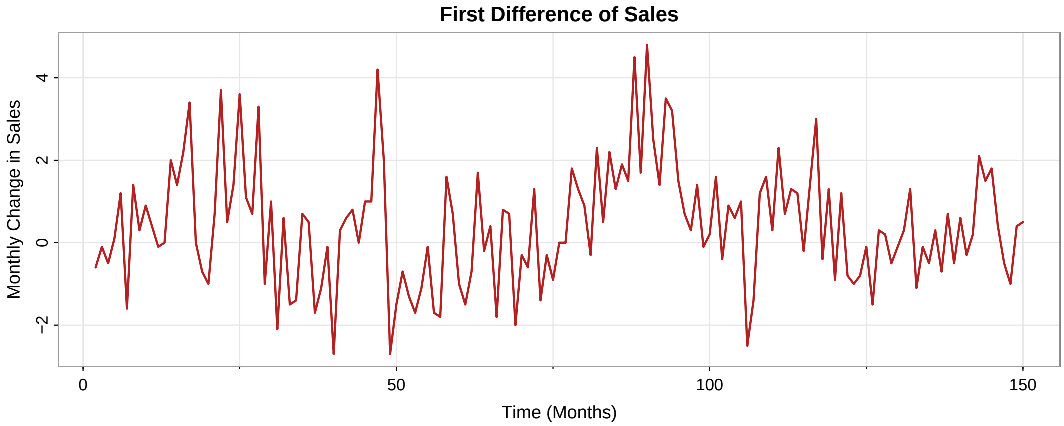
Problem 1 (b)

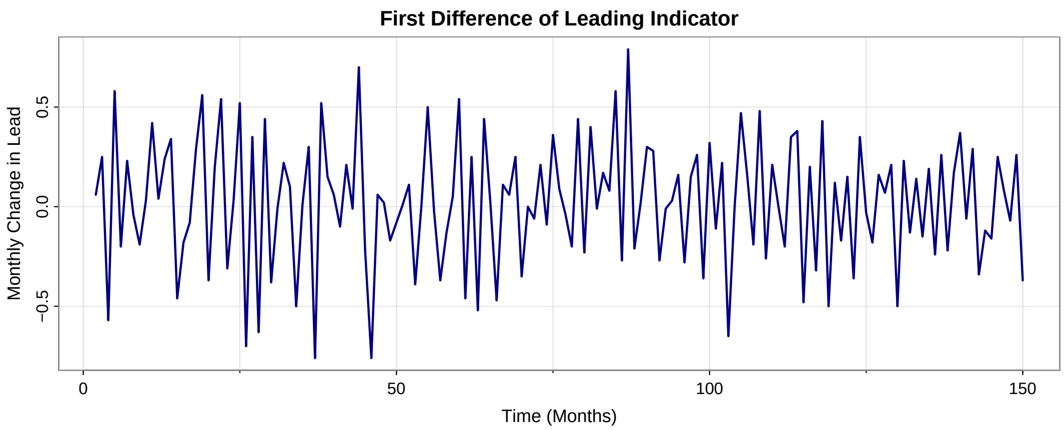
Compute first ordered differenced series for both `sales` and `lead`. Show plots of the differenced time series and discuss their main features.

Cell 3

```r
# Build first differences.
sales.diff <- diff(sales)
lead.diff  <- diff(lead)

# Plot the differenced series.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 1))
astsa::tsplot(
  sales.diff,
  main = 'First Difference of Sales',
  xlab = 'Time (Months)',
  ylab = 'Monthly Change in Sales',
  col = 'firebrick',
  lwd = 2
)
astsa::tsplot(
  lead.diff,
  main = 'First Difference of Leading Indicator',
  xlab = 'Time (Months)',
  ylab = 'Monthly Change in Lead',
  col = 'navy',
  lwd = 2
)
```





Let $U_t = \nabla S_t = S_t - S_{t-1}$ and $V_t = \nabla L_t = L_t - L_{t-1}$. After differencing, both series have 149 observations. The differenced sales series fluctuates around a much more stable level than the original sales series, and the differenced lead series does the same for the leading indicator. In both plots, the major trend has largely been removed, so the mean is much less time-dependent than in part (a). There is still no clear repeated seasonal pattern. The statistical ranges are $\max(U_t) - \min(U_t) = 7.5$ and $\max(V_t) - \min(V_t) = 1.55$. The spread looks roughly constant over time, and although there are a few larger jumps, the differenced series are much less variable and much less smooth than the original series. The differenced series also do not look especially volatile; instead, they behave like fairly stable fluctuations around zero with occasional isolated spikes, with U_t showing the larger jumps and V_t staying within a narrower band.

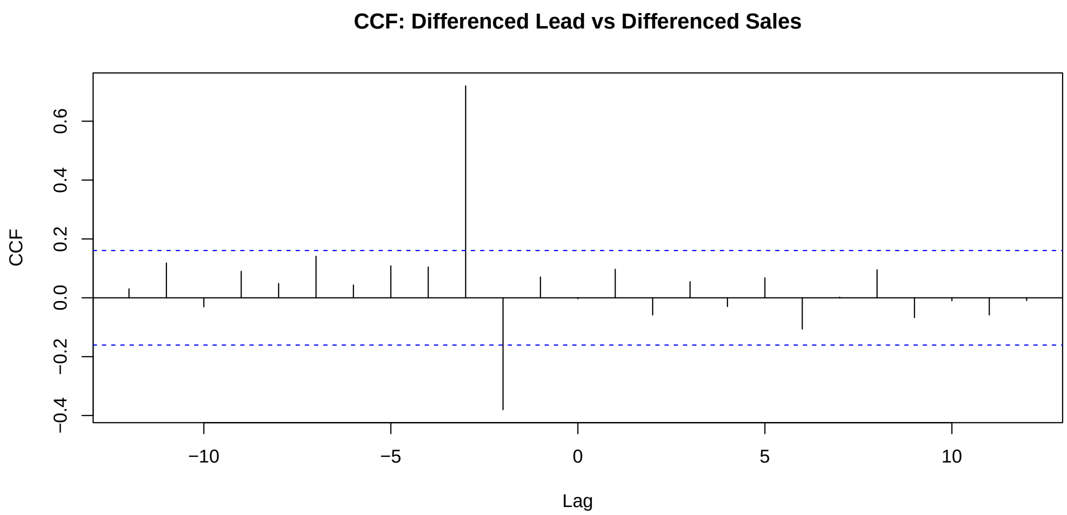
Problem 1 (c)

Problem 1 (c)

Plot the CCF of first ordered differenced `lead` and first ordered differenced `sales`. Comment on the relation between the two differenced series in specific lead-lag terminology learned in class.

Cell 4

```r
# Plot CCF of differenced series.
cc.diff <- stats::ccf(
  as.numeric(lead.diff),
  as.numeric(sales.diff),
  lag.max = 12,
  main = 'CCF: Differenced Lead vs Differenced Sales',
  ylab = 'CCF'
)
```



Cell 5

```r
# Extract dominant lag.
cc.max.idx <- which.max(abs(cc.diff$acf))
cc.max.lag <- as.numeric(cc.diff$lag[cc.max.idx])
cc.max.acf <- as.numeric(cc.diff$acf[cc.max.idx])
```

The dominant CCF spike occurs at lag -3 with cross-correlation 0.72. Since the first series passed into `ccf()` is `lead.diff`, a negative lag means the differenced leading indicator leads the differenced sales series. Therefore, the CCF indicates that V_t leads U_t by 3 months, or equivalently, U_t lags V_t by 3 months. The relation is positive at that lag, so increases in the differenced leading indicator tend to be followed by increases in differenced sales about 3 months later.

Problem 1 (d)

Problem 1 (d)

Fit the regression model

$$\nabla S_t = \beta_0 + \beta_1 \nabla L_{t-3} + X_t,$$

where X_t is the residual process after performing regression. Present the fitted model for $U_t = \nabla S_t$ and discuss how it performed using the details from the summary of this model.

Cell 6

```r
# Fit the lag-3 regression suggested by the CCF.
DiffData <- stats::ts.intersect(U = sales.diff, Lt3 = stats::lag(lead.diff, -3),
dframe = TRUE)
fit <- stats::lm(U ~ Lt3, data = DiffData)

# Summarize model.
summary(fit)
```

```
##
## Call:
## stats::lm(formula = U ~ Lt3, data = DiffData)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.18876 -0.65502 -0.07291  0.60347  2.84546
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.35538    0.08301   4.281            0.0000338 ***
## Lt3          3.33733    0.26190  12.743 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1 on 144 degrees of freedom
## Multiple R-squared:   0.53,  Adjusted R-squared:  0.5267
## F-statistic: 162.4 on 1 and 144 DF,  p-value: < 0.00000000000000022
```

The fitted model for $U_t = \nabla S_t$ is

$$\widehat{U}_t = 0.355382 + 3.33733 V_{t-3},$$

where $U_t = \nabla S_t$ is the monthly change in sales and $V_t = \nabla L_t$ is the monthly change in the leading indicator. After aligning U_t with V_{t-3}, the fitted sample contains $n = 146$ observations.

- $\hat{\beta}_0 = 0.355$ means that when the differenced leading indicator 3 months earlier is 0, the expected current monthly change in sales is about 0.355 sales units.
- $\hat{\beta}_1 = 3.337$ means that a 1-unit increase in the differenced leading indicator 3 months earlier is associated with about 3.337 units increase in the current monthly change in sales.

This matches the positive lead-lag relation seen in the CCF. From the model summary, the predictor is highly significant ($p < 2 \times 10^{-16}$), and the intercept is also significant ($p = 0.00003377$). The coefficient of determination is $R^2 = 0.53$, so the model explains about 53% of the variation in the monthly change in sales. The residual standard error is 1 in sales-difference units, which means on average, the predicted current monthly change in sales deviates from the actual current monthly change in sales by about 1 units. So, from the summary, the lag-3 predictor performs strongly as an explanatory variable for the differenced sales series, and the model is decent.

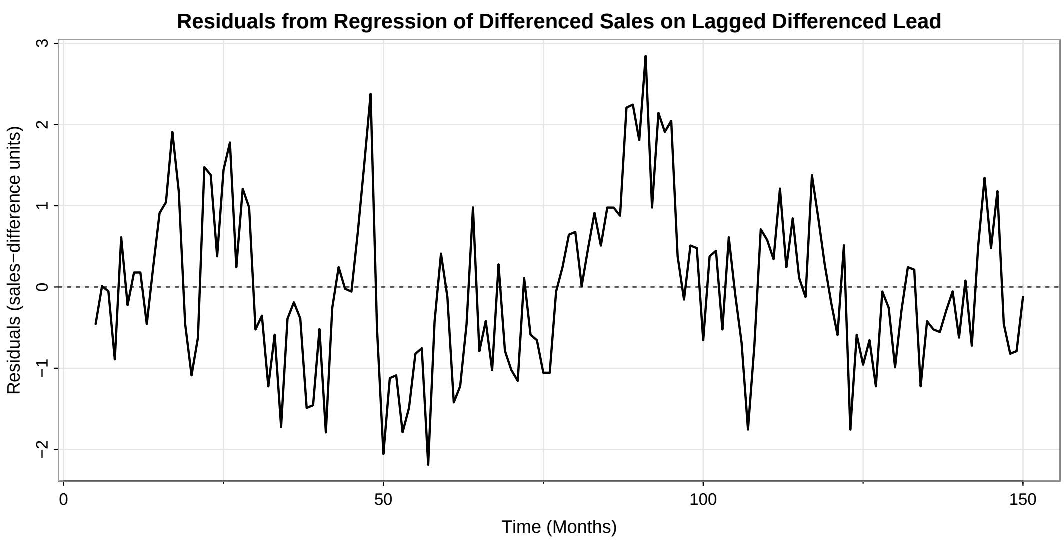
Problem 1 (e)

Problem 1 (e)

For the above model, discuss the goodness of fit using coefficient of determination, statistical significance of predictor and model, residual standard error and residual plot.

Cell 7

```
# Plot residual process.
res.ts <- stats::ts(stats::residuals(fit), start = 5, frequency = 1)
astsa::tsplot(
  res.ts,
  main = 'Residuals from Regression of Differenced Sales on Lagged Differenced Lead',
  xlab = 'Time (Months)',
  ylab = 'Residuals (sales-difference units)',
  lwd = 2
)
graphics::abline(h = 0, lty = 2)
```



While some may be redundant with my response in part (d), the predictor is highly significant ($p < 2 \times 10^{-16}$), and the overall regression model is also highly significant because the F-test has $p < 2.2 \times 10^{-16}$. The coefficient of determination is $R^2 = 0.53$, so the model explains about 53% of the variation in the monthly change in sales. The residual standard error is 1 in sales-difference units. This means on average, the predicted current monthly change in sales deviates from the actual current monthly change in sales by about 1 units. From the residual plot, the residuals fluctuate around zero, but they do not look completely random. There are noticeable runs of positive and negative residuals and some clustering, so the regression captures an important lagged relation but does not fully remove all dependence in the differenced sales series. Overall, the model performs reasonably well in terms of significance and R^2, but the residual plot suggests that it is not the full story.

Problem 2

Problem 2

For your data analysis project, do the following.

Cell 8

```r
# Load the curated data project files.
blackwood_gbgs <- utils::read.csv(
  'data/blackwood_gbgs.csv',
  stringsAsFactors = FALSE
)
shesterkin_gbgs <- utils::read.csv(
  'data/shesterkin_gbgs.csv',
  stringsAsFactors = FALSE
)
```

Problem 2 (a)

Problem 2 (a)

Introduce your data project problem in detail explaining research goals clearly. Share your data in detail with all variables, source, and how this data was selected (that is, why not another dataset).

My data project asks whether a goalie's goals saved above expected, or GSAx, is really as team-independent as it is often treated in hockey analysis. The usual appeal of GSAx is that it seems fairer than raw save percentage or goals against because it does not treat all shots as equal; instead, it rewards a goalie for stopping dangerous chances and penalizes him more for allowing easier ones, so in principle it already adjusts for the playing conditions his team gives him. I want to challenge that interpretation rather than simply accept it. More specifically, I want to compare a goalie whose team environment changed substantially with a goalie whose team environment stayed much more stable. Mackenzie Blackwood is useful because he has now played for New Jersey, San Jose, and Colorado, so his surrounding roster and team process have changed a great deal. Igor Shesterkin is useful as the contrast because he has stayed with the New York Rangers, so any changes in his series come from year-to-year roster evolution rather than a franchise change. At the same time, I do not want to overstate the claim, because goalie performance can also vary for reasons that are not fully captured by team context, including natural progression or regression over a career, injury or recovery, confidence and mental state, fatigue, or back-to-back scheduling. So the project is not asking whether team context is the only driver of GSAx; it is asking whether the behavior of GSAx looks different when one goalie experiences major team changes and another does not.

I study both goalies game by game at even strength and keep only regular-season games, that is, games with `gameTypeId = 2`, equivalently `gameId` values whose middle two digits are `02`. After filtering to complete even-strength regular-season records, Blackwood contributes the smaller complete sample, so to keep the comparison balanced I restrict both goalies to the first 285 complete even-strength regular-season career games. For Blackwood, the analysis sample runs from 2018-12-18 through 2026-03-28 and includes 152 games with New Jersey, 63 games with San Jose, and 70 games with Colorado. For Shesterkin, the matched sample runs from 2020-01-07 through 2025-11-04 and all 285 games are with the Rangers. Within each goalie's file, the time index is career game number, so for Blackwood and Shesterkin separately I set $t = 1, 2, \ldots, 285$.

The raw NHL event and report data in my broader pipeline do start with my `nhlscraper` package through functions such as `nhlscraper::games()`, `nhlscraper::goalie_game_report()`, `nhlscraper::gc_pbps()`, and `nhlscraper::shift_charts()`, but the expected-goals layer used here is not merely a direct `nhlscraper` output. In `rentosrink`, I build my own xG pipeline on top of those inputs. The current model and deployment logic are described in the live Streamlit model page at `rentosrink.streamlit.app` and in the source file `rentosrink/views/expected_goal.py`. The advanced game-by-game files used here are described in the GBG codebook at `rentosrink/scripts/gbgs/codebook.md`, which states that advanced GBGs are built from the SBS/xG pipeline rather than from raw reports alone.

Because this project is restricted to even-strength data, the relevant 2025–26 unseen-future xG validation is the two even-strength partitions rather than an all-situations aggregate. For Standard 5v5, the deployed `sd1` XGBoost model achieved log loss = 0.1926, Brier score = 0.0522, ROC AUC = 0.8038, and calibration ratio = 1.0477. For Non-Standard Even Strength, the deployed `ev2` LightGBM model achieved log loss = 0.2986, Brier score = 0.0853, ROC AUC = 0.7365, and calibration ratio = 1.1595. Those numbers matter here because my project depends on the quality of the xG estimates that feed the even-strength variables `xga_ev`, `xgf_ev`, `gsax_ev`, and `team_xg_pct_ev`; if the xG model were poor, then the entire goalie-context question would be much weaker. At the same time, even a strong public xG model is still incomplete. In particular, the public NHL event data do not provide full pass-tracking information, so my public xG models still cannot fully account for detailed pre-shot passing structure, deception, or some off-puck context. That means if GSAx still behaves differently across the two goalie situations, the project may reveal not only genuine team influence on goalie results, but also what the current public xG framework is still missing.

The final project files were created by filtering the `rentosrink` goalie and team game-by-game tables in `rentosrink/data/gbgs/basic` and `rentosrink/data/gbgs/advanced` to Blackwood's player id `8478406` and Shesterkin's player id `8478048`, keeping only regular-season games, merging the goalie basic, goalie advanced, and team advanced files by game and team, joining team names from `rentosrink/data/teams.csv`, and then keeping the even-strength columns only. The final analysis files are `STA209/data/blackwood_gbgs.csv` and `STA209/data/shesterkin_gbgs.csv`, and each contains 285 complete observations after matching the sample length.

The identifier and indexing variables are `game_number`, which is the time index t; `game_date`, the calendar date of the game; `season`, the NHL season id; `team_id`, `team_abbr`, and `team_name`, which identify the goalie's team in that game; `game_id` and `player_id`; and `team_change`, an indicator for the first game after a franchise change. The hockey performance variables are `shots_against_ev`, `goals_against_ev`, `xga_ev`, `xgf_ev`, `sv_pct_ev`, `gsax_ev`, `cumulative_gsax_ev`, and `cumulative_sv_pct_ev`. The main derived team-context variable is

$$\texttt{team\_xg\_pct\_ev} = 100 \times \frac{\texttt{xgf\_ev}}{\texttt{xgf\_ev} + \texttt{xga\_team\_ev}},$$

which is the team even-strength expected-goal share percentage from the team GBG file. For the actual analysis in part (c), I use `cumulative_gsax_ev` as the main response series and also model `gsax_ev`, `sv_pct_ev`, and `team_xg_pct_ev` separately for each goalie, since the assignment asks for modeling at least three time series and these four series together describe both goalie results and team environment.

Problem 2 (b)

Problem 2 (b)

Find three journal articles on the research topic of your interest (called background research) using Google Scholar and OneSearch to support the relevance and importance of the research questions you are exploring. Cite the background research you did and share a summary.

Using Google Scholar and OneSearch, I focused on journal articles that support three separate parts of my project question: whether hockey scoring environments are context-dependent, whether expected-goals style process metrics are more informative than raw results, and whether goalkeeper-specific shot models should explicitly account for the keeper rather than treating all context as external.

- The first article is Lignell, Rago, and Mohr (2020), "Analysis of goal scoring opportunities in elite male ice hockey in relation to tactical and contextual variables," *International Journal of Performance Analysis in Sport*, 20(6), 1003–1017, https://doi.org/10.1080/24748668.2020.1823161. This article is directly relevant because it is hockey-specific and shows that the probability of scoring is affected by tactical and contextual variables, not just by a shooter-goalie duel in isolation, which supports the idea that a goalie's environment may matter when we interpret a metric like GSAx.
- The second article is Mead, O'Hare, and McMenemy (2023), "Expected goals in football: Improving model performance and demonstrating value," *PLOS ONE*, 18(4), e0282295, https://doi.org/10.1371/journal.pone.0282295. Even though it is soccer rather than hockey, it is still very relevant because it shows that expected-goals style process measures outperform traditional shot and goal counts for prediction and evaluation, which supports using xG-based context variables instead of relying only on goals against or wins.
- The third article is De-la-Cruz-Torres, Navarro-Castro, and Ruiz-de-Alarcón-Quintero (2025), "An Expected Goals On Target (xGOT) Model: Accounting for Goalkeeper Performance in Football," *Big Data and Cognitive Computing*, 9(3), 64, https://doi.org/10.3390/bdcc9030064. This article matters because it argues that traditional shot models often overlook the goalkeeper's contribution and proposes an explicitly goalkeeper-aware post-shot model, which is conceptually close to my question because GSAx is also supposed to isolate goalie performance from chance quality.

All in all, these three articles support the importance of my project from three directions: hockey chances are shaped by context, xG-type process metrics are valuable for performance evaluation, and goalkeeper evaluation is strongest when the model takes both chance quality and the keeper's contribution seriously rather than assuming independence by default. They also justify the comparison between Blackwood and Shesterkin, because that design lets me look at one goalie under major team changes and another under relative organizational continuity without leaving the broader question of context-adjusted goalie evaluation.

Problem 2 (c)

Problem 2 (c)

For each variable, show time series plots (you can do multiple plots on the same slide) and comment on its main features (trend, seasonality, range, volatile, heteroscedasticity). Model the trend and seasonality of all variables (at least three) using techniques learned in the class and report the best model justifying their performance using all methods learned in class. Make a summary table comparing adjusted R^2, F-test, likelihood-based criteria, and forecast evaluation criteria. For seasonality, make a periodogram of your response variable (one time series) and see if there is any seasonality in the data.

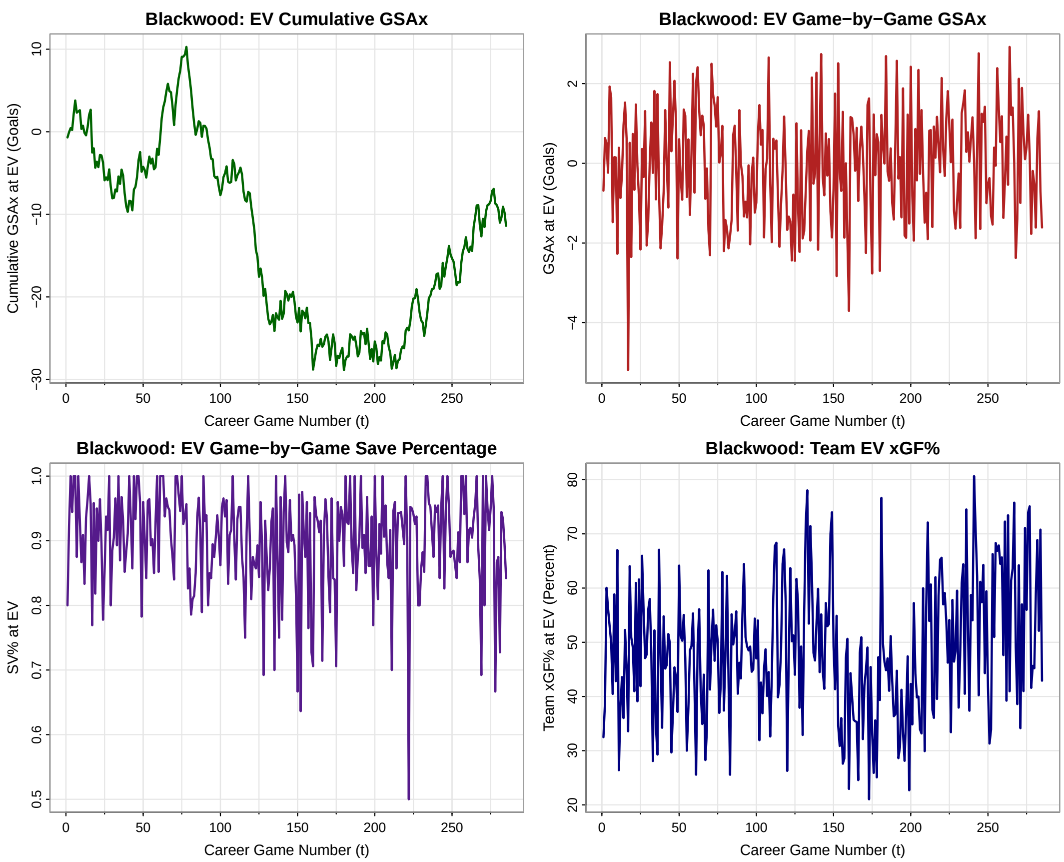
For this part, I treat each series as a univariate time series indexed by career game number. For both Blackwood and Shesterkin, the time variable is $t = 1, 2, \ldots, 285$, where each value of t corresponds to one complete even-strength game in that goalie's matched sample. The four series are `cumulative_gsax_ev`, `gsax_ev`, `sv_pct_ev`, and `team_xg_pct_ev`. For forecast evaluation, I use Blackwood's move to Colorado as the split point, so the training sample is $t = 1, \ldots, 215$ and the holdout sample is $t = 216, \ldots, 285$. For Blackwood, that means training through 2024-12-07 and forecasting from 2024-12-14, his first Colorado game. For Shesterkin, I keep the same career-game split $t = 216$ so that the holdout period has the same length and the comparison stays aligned by career-game count rather than calendar date.

Mackenzie Blackwood

Cell 9

```r
# Time series plots for Blackwood.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 2))
astsa::tsplot(
  stats::ts(blackwood_gbgs$cumulative_gsax_ev, start = 1, frequency = 1),
  main = 'Blackwood: EV Cumulative GSAx',
  xlab = 'Career Game Number (t)',
  ylab = 'Cumulative GSAx at EV (Goals)',
  col = 'darkgreen',
  lwd = 2
)
astsa::tsplot(
  stats::ts(blackwood_gbgs$gsax_ev, start = 1, frequency = 1),
  main = 'Blackwood: EV Game-by-Game GSAx',
  xlab = 'Career Game Number (t)',
  ylab = 'GSAx at EV (Goals)',
  col = 'firebrick',
  lwd = 2
)
astsa::tsplot(
  stats::ts(blackwood_gbgs$sv_pct_ev, start = 1, frequency = 1),
  main = 'Blackwood: EV Game-by-Game Save Percentage',
```

```
  xlab = 'Career Game Number (t)',
  ylab = 'SV% at EV',
  col = 'purple4',
  lwd = 2
)
astsa::tsplot(
  stats::ts(blackwood_gbgs$team_xg_pct_ev, start = 1, frequency = 1),
  main = 'Blackwood: Team EV xGF%',
  xlab = 'Career Game Number (t)',
  ylab = 'Team xGF% at EV (Percent)',
  col = 'navy',
  lwd = 2
)
```



The `cumulative_gsax_ev` data is Blackwood's running even-strength goals saved above expected in goals over 285 career games from 2018-12-18 through 2026-03-28. Its mean behavior is clearly time-dependent and non-linear: the series rises early, declines for a long middle stretch, and then partially recovers late, which matches the idea that Blackwood experienced meaningful changes in team and career context over time. There is no visible repeated seasonal pattern. Its statistical range is $\max(CG_t) - \min(CG_t) = 39.145$ goals. Because the series is cumulative, the spread changes gradually rather than abruptly, so there is no strong evidence of heteroscedastic fanning or short-run volatility clustering.

The `gsax_ev` data is Blackwood's game-by-game even-strength GSAx in goals over the same 285 games. Unlike the cumulative series, it fluctuates around a fairly stable center and does not show a clear deterministic long-run trend, although there are clusters of better and worse games. There is again no visible repeated seasonal pattern. Its statistical range is $\max(G_t) - \min(G_t) = 8.111$ goals. The variance looks fairly stable over time, so it appears approximately homoskedastic, but there are several isolated positive and negative spikes, so the series is clearly volatile in the short run.

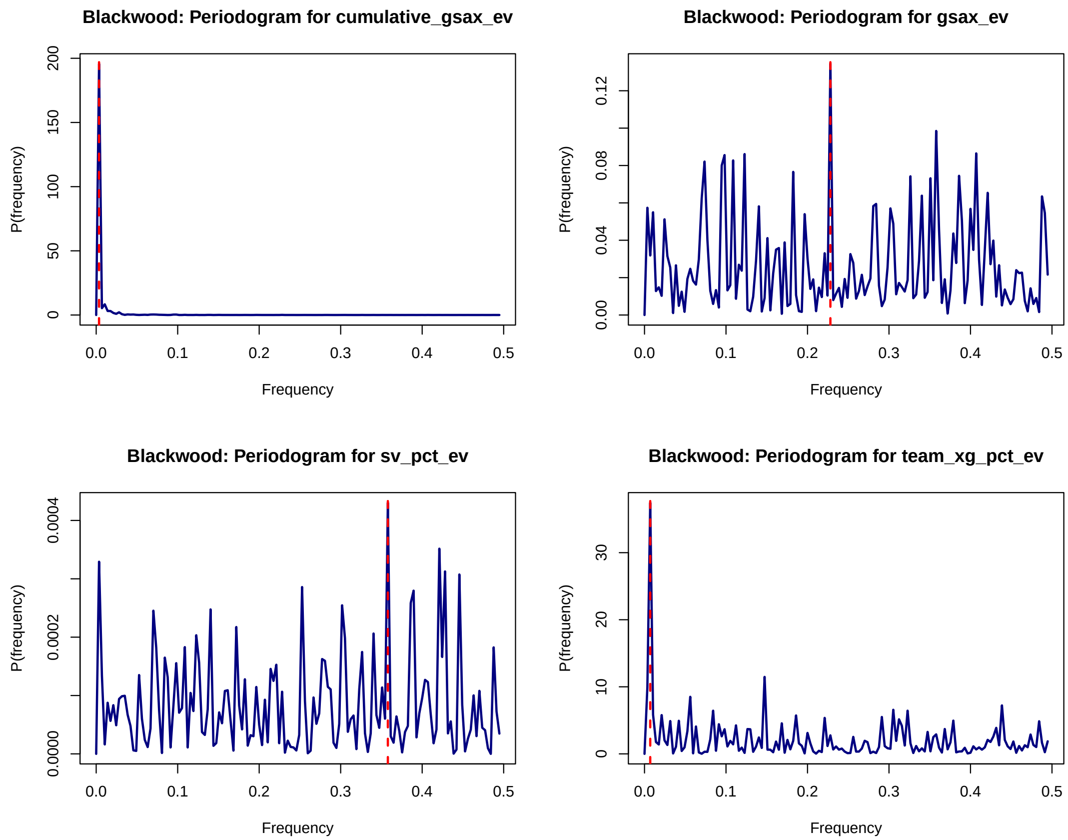
The `sv_pct_ev` data is Blackwood's game-by-game even-strength save percentage in proportion form. The mean appears more stable than the cumulative GSAx series, though there is still mild curvature over the full career path rather than a perfectly flat level. There is no clear repeated seasonal pattern. Its statistical range is $\max(S_t) - \min(S_t) = 0.5$. The spread looks broadly stable over time, so there is no strong visual evidence of heteroscedasticity, but individual games can still jump sharply up or down, which makes the series moderately volatile.

The `team_xg_pct_ev` data is Blackwood's team even-strength expected-goal share percentage in percentage points. Its mean behavior is time-dependent, but not in a smooth linear way; instead, it shows broader regime changes that line up naturally with changes in team environment. There is no visible repeated seasonal pattern. Its statistical range is $\max(X_t) - \min(X_t) = 59.627$ percentage points. The variance appears roughly stable over time, so the series looks approximately homoskedastic, and while it does move materially, it is not as spike-like or volatile as the two game-by-game goalie-performance series.

Cell 10

```r
# Periodograms for Blackwood.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 2))
for (nm in c('cumulative_gsax_ev', 'gsax_ev', 'sv_pct_ev', 'team_xg_pct_ev')) {
  info <- blackwood_analysis[[nm]]$period
  title_text <- switch(
    nm,
    cumulative_gsax_ev = 'Blackwood: Periodogram for cumulative_gsax_ev',
    gsax_ev = 'Blackwood: Periodogram for gsax_ev',
    sv_pct_ev = 'Blackwood: Periodogram for sv_pct_ev',
    team_xg_pct_ev = 'Blackwood: Periodogram for team_xg_pct_ev'
  )
  graphics::plot(
    info$f,
    info$P,
    type = 'l',
    main = title_text,
    xlab = 'Frequency',
    ylab = 'P(frequency)',
    col = 'navy',
    lwd = 2
  )
  graphics::abline(v = info$freq, col = 'red', lty = 2, lwd = 2)
```

```
}
```



For Blackwood, none of the four periodograms suggests a convincing seasonal cycle. For `cumulative_gsax_ev`, there is a very prominent spike, but it is concentrated essentially at frequency 0, with dominant frequency $f = 0.003509$ and implied period about 285 games, so that spike reflects long-run trend rather than seasonality. For `gsax_ev` and `sv_pct_ev`, the highest peaks occur at implied periods of about 4.38 and 2.79 games, but those peaks are not sharply separated from many other nearby peaks of similar height, so the periodograms do not show the kind of isolated dominant spike that would support a clean seasonal frequency. For `team_xg_pct_ev`, the stronger low-frequency peak again sits very close to 0 and is more consistent with broad team-context drift or regime change than with a stable repeating cycle. Therefore, I do not include harmonic terms for Blackwood.

Since no meaningful seasonality is present, I compare the same four nonseasonal polynomial trend models for each Blackwood series Y_t:

$$\text{Model I: } Y_t = \beta_0 + \beta_1 t + W_t,$$
$$\text{Model II: } Y_t = \beta_0 + \beta_1 t + \beta_2 \frac{t^2}{2!} + W_t,$$
$$\text{Model III: } Y_t = \beta_0 + \beta_1 t + \beta_2 \frac{t^2}{2!} + \beta_3 \frac{t^3}{3!} + W_t,$$
$$\text{Model IV: } Y_t = \beta_0 + \beta_1 t + \beta_2 \frac{t^2}{2!} + \beta_3 \frac{t^3}{3!} + \beta_4 \frac{t^4}{4!} + W_t.$$

Blackwood: Cumulative Goals Saved Above Expected

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.4058	<0.0001	7.0666	5.2290	7.1050
Model II	3	0.5716	<0.0001	6.7429	4.9055	6.7941
Model III	4	0.8185	<0.0001	5.8874	4.0503	5.9515
Model IV	5	0.8223	<0.0001	5.8700	4.0332	5.9469

Model	ME	MPE	MSE	MAE	MAPE
Model I	18.5387	-152.0291	418.3032	18.5387	152.0291
Model II	30.4078	-245.2769	1084.0513	30.4078	245.2769
Model III	-0.4547	11.8134	24.6064	3.7304	29.7550
Model IV	-64.6335	572.4741	6653.5218	64.6335	572.4741

Blackwood: Game-by-Game GSAx

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.0004	0.2898	3.4953	1.6577	3.5337
Model II	3	0.0019	0.2839	3.4973	1.6599	3.5486
Model III	4	-0.0006	0.4180	3.5032	1.6661	3.5673
Model IV	5	0.0106	0.1375	3.4955	1.6587	3.5724

Model	ME	MPE	MSE	MAE	MAPE
Model I	0.4424	136.2439	1.6950	1.0868	144.0853
Model II	0.0921	80.5572	1.5588	1.0319	114.2257
Model III	-0.8172	-55.4203	2.7270	1.3039	341.7306
Model IV	2.0009	343.5387	6.9712	2.1541	578.1639

Blackwood: Game-by-Game Save Percentage

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.0023	0.1980	-2.2437	-4.0813	-2.2053
Model II	3	0.0128	0.0598	-2.2508	-4.0882	-2.1995
Model III	4	0.0094	0.1300	-2.2439	-4.0810	-2.1798
Model IV	5	0.0264	0.0215	-2.2577	-4.0945	-2.1808

Model	ME	MPE	MSE	MAE	MAPE
Model I	0.0329	2.4804	0.0085	0.0713	8.3349
Model II	0.0043	-0.7235	0.0075	0.0630	7.6539
Model III	-0.0894	-11.2195	0.0192	0.0999	12.2918
Model IV	-0.0254	-4.0329	0.0082	0.0613	7.6986

Blackwood: Team xGF%

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.0275	0.0029	7.8687	6.0312	7.9072
Model II	3	0.0584	<0.0001	7.8399	6.0026	7.8912
Model III	4	0.0742	<0.0001	7.8265	5.9894	7.8906
Model IV	5	0.0709	<0.0001	7.8335	5.9967	7.9104

Model	ME	MPE	MSE	MAE	MAPE
Model I	12.8374	18.6796	319.6367	14.8493	24.3470
Model II	20.3512	32.9025	582.9084	20.6871	33.9199
Model III	22.7150	37.3670	696.3286	22.8995	37.9302
Model IV	-36.6940	-74.6255	2858.3043	40.1725	80.0491

For Blackwood's `cumulative_gsax_ev`, Model IV has the largest adjusted R^2 and the smallest AIC, AICc, and BIC, but once the holdout period is defined by the Colorado move, its forecast performance deteriorates badly. Since `cumulative_gsax_ev` can be negative or near zero, I rely more on MSE and MAE than on MPE and MAPE for forecast evaluation here. Model III keeps essentially all of the strong in-sample fit while forecasting much better across the Colorado segment, so I select Model III as the practical final model for Blackwood's cumulative GSAx. The fitted model is

$$\widehat{CG}_t = -5.460662 + 0.305395t - 0.008922\frac{t^2}{2!} + 0.000072\frac{t^3}{3!}$$

where CG_t is Blackwood's cumulative even-strength GSAx in goals at career game $t = 1, 2, \ldots, 285$.

For Blackwood's `gsax_ev`, the story is very different. All four models have adjusted R^2 values near zero and the overall F-tests are not significant, so there is no strong evidence of a meaningful deterministic polynomial trend in Blackwood's game-by-game GSAx. Under the Colorado split, however, the holdout MSE and MAE are both smallest for Model II, even though the information criteria still prefer Model I. Since the forecasting target here is specifically the post-trade segment and Model II is only one step more complex, I select Model II as the better practical forecasting model. The fitted model is

$$\widehat{G}_t = 0.027957 - 0.003516t + 0.000032\frac{t^2}{2!}$$

where G_t is Blackwood's even-strength game-by-game GSAx in goals at career game t.

For Blackwood's `sv_pct_ev`, the criteria remain mixed, but the Colorado split shifts the forecasting preference toward the quadratic model. Model IV still gives the largest adjusted

R^2 and the smallest AIC and AICc, but Model II gives the smallest holdout MSE, while Model IV only narrowly improves MAE. Since the quadratic model is much simpler and already has a significant overall F-test, I select Model II as the more reasonable compromise between fit, forecasting, and parsimony. The fitted model is

$$\widehat{S}_t = 0.933529 - 0.000509t + 0.000003\frac{t^2}{2!}$$

where S_t is Blackwood's even-strength save percentage in proportion form at career game t.

For Blackwood's `team_xg_pct_ev`, the Colorado split changes the conclusion materially. Model III still has the largest adjusted R^2 and the smallest AIC, AICc, and BIC, but when the forecasting target is the post-trade Colorado segment, Model I gives the smallest holdout MSE and MAE by a wide margin. Since the team change creates a real regime break, I place more weight on forecast performance here than on the full-sample descriptive fit, and I select Model I as the more useful model for this exercise. The fitted model is

$$\widehat{X}_t = 44.788655 + 0.026593t$$

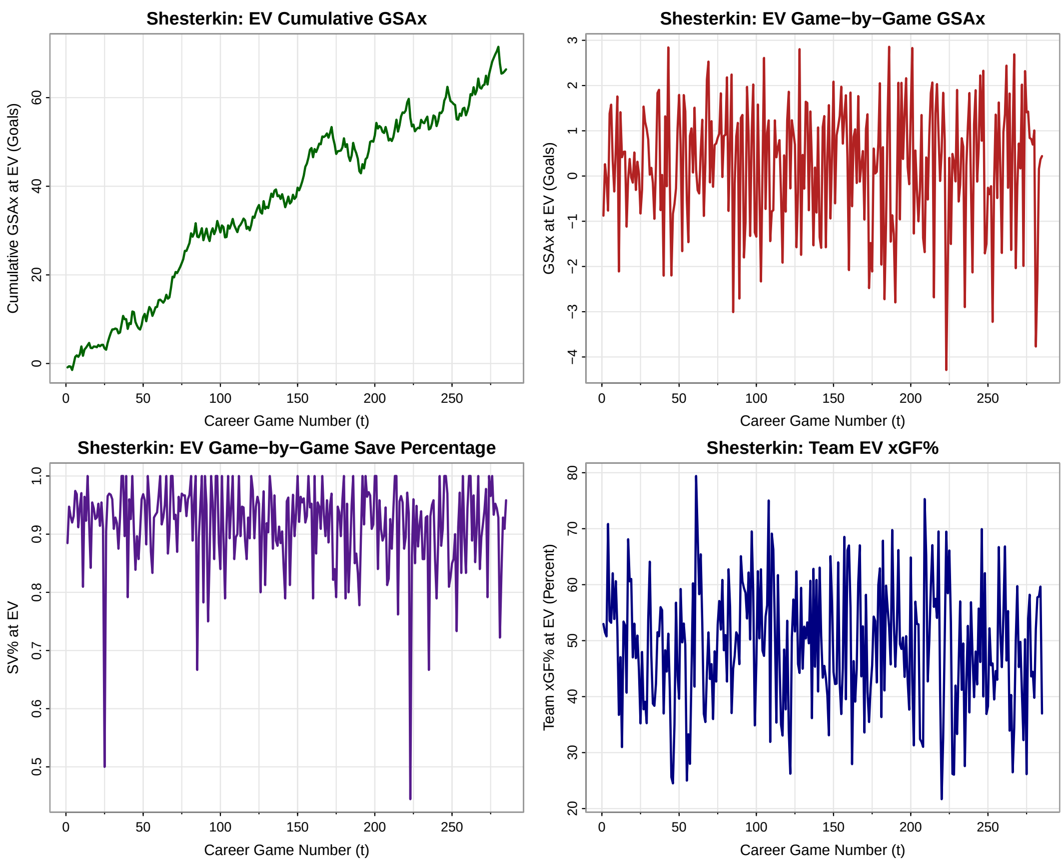
where X_t is Blackwood's team even-strength xGF% in percentage points at career game t.

Igor Shesterkin

Cell 11

```r
# Time series plots for Shesterkin.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 2))
astsa::tsplot(
  stats::ts(shesterkin_gbgs$cumulative_gsax_ev, start = 1, frequency = 1),
  main = 'Shesterkin: EV Cumulative GSAx',
  xlab = 'Career Game Number (t)',
  ylab = 'Cumulative GSAx at EV (Goals)',
  col = 'darkgreen',
  lwd = 2
)
astsa::tsplot(
  stats::ts(shesterkin_gbgs$gsax_ev, start = 1, frequency = 1),
  main = 'Shesterkin: EV Game-by-Game GSAx',
  xlab = 'Career Game Number (t)',
  ylab = 'GSAx at EV (Goals)',
  col = 'firebrick',
  lwd = 2
)
astsa::tsplot(
  stats::ts(shesterkin_gbgs$sv_pct_ev, start = 1, frequency = 1),
  main = 'Shesterkin: EV Game-by-Game Save Percentage',
  xlab = 'Career Game Number (t)',
  ylab = 'SV% at EV',
  col = 'purple4',
  lwd = 2
)
```

```r
astsa::tsplot(
  stats::ts(shesterkin_gbgs$team_xg_pct_ev, start = 1, frequency = 1),
  main = 'Shesterkin: Team EV xGF%',
  xlab = 'Career Game Number (t)',
  ylab = 'Team xGF% at EV (Percent)',
  col = 'navy',
  lwd = 2
)
```



The `cumulative_gsax_ev` data is Shesterkin's running even-strength goals saved above expected in goals over his first 285 matched career games from 2020-01-07 through 2025-11-04. Compared with Blackwood, the most striking difference is that Shesterkin's cumulative GSAx follows a much steadier upward path and finishes far above zero. The mean behavior is clearly time-dependent and non-linear, but the dominant shape is strong upward growth with only moderate flattening or pullback episodes. There is no visible repeated seasonal pattern. Its statistical range is $\max(CG_t) - \min(CG_t) = 72.932$ goals. As with Blackwood, the cumulative nature of the series means there is no obvious short-run volatility clustering or abrupt fanning of variance.

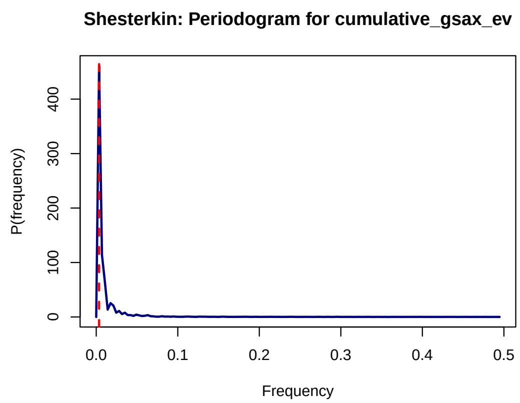
The `gsax_ev` data is Shesterkin's game-by-game even-strength GSAx in goals. Compared with Blackwood's game-by-game GSAx, it fluctuates around a somewhat higher center, which is consistent with Shesterkin's much more positive cumulative path, but it is still a noisy game-by-game series with no strong deterministic trend. There is no visible repeated seasonal pattern. Its statistical range is $\max(G_t) - \min(G_t) = 7.145$ goals. The spread looks fairly stable over time, so the series appears approximately homoskedastic, and there are isolated positive and negative spikes, so it remains volatile in the short run.

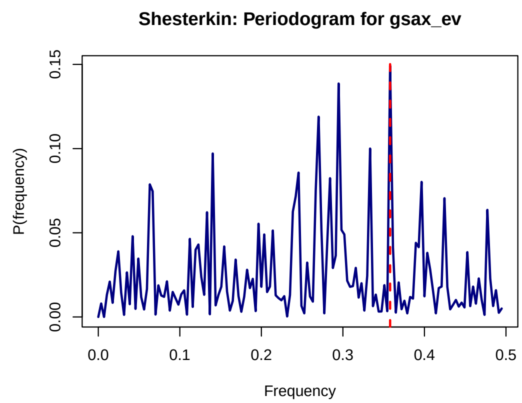
The `sv_pct_ev` data is Shesterkin's game-by-game even-strength save percentage in proportion form. Compared with Blackwood, Shesterkin's series sits at a slightly higher average level and appears a little more stable overall, although it still has occasional sharp one-game deviations. There is no visible repeated seasonal pattern. Its statistical range is $\max(S_t) - \min(S_t) = 0.5556$. The variance looks roughly constant across time, so there is no strong visual sign of heteroscedasticity, and the short-run volatility is moderate rather than extreme.

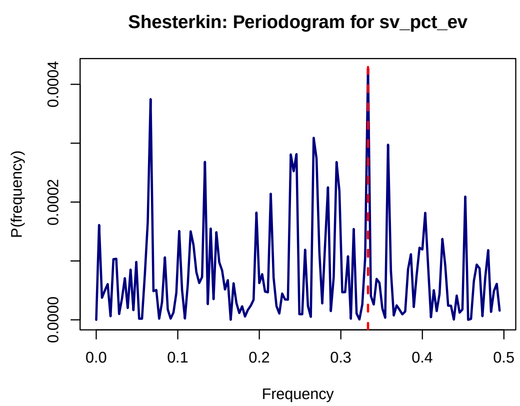
The `team_xg_pct_ev` data is Shesterkin's team even-strength expected-goal share percentage in percentage points over the same 285 games. Compared with Blackwood, this series looks less dominated by franchise-level regime changes because Shesterkin stays with one organization throughout the matched sample. Its mean behavior still moves over time, but the shifts are smoother and more incremental. There is no visible repeated seasonal pattern. Its statistical range is $\max(X_t) - \min(X_t) = 57.737$ percentage points. The spread appears roughly constant over time, so the series looks approximately homoskedastic, and it is less spiky than the goalie-performance series.

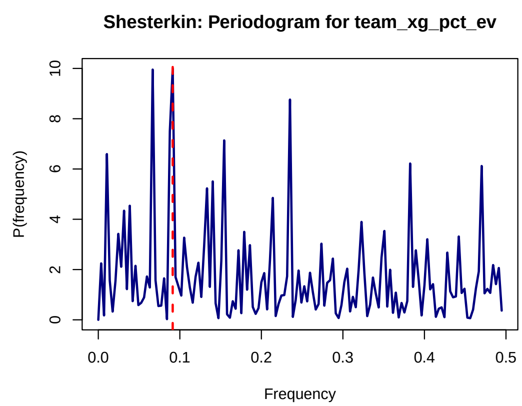
Cell 12

```r
# Periodograms for Shesterkin.
op <- graphics::par(no.readonly = TRUE)
on.exit(graphics::par(op), add = TRUE)
graphics::par(mfrow = c(2, 2))
for (nm in c('cumulative_gsax_ev', 'gsax_ev', 'sv_pct_ev', 'team_xg_pct_ev')) {
  info <- shesterkin_analysis[[nm]]$period
  title_text <- switch(
    nm,
    cumulative_gsax_ev = 'Shesterkin: Periodogram for cumulative_gsax_ev',
    gsax_ev = 'Shesterkin: Periodogram for gsax_ev',
    sv_pct_ev = 'Shesterkin: Periodogram for sv_pct_ev',
    team_xg_pct_ev = 'Shesterkin: Periodogram for team_xg_pct_ev'
  )
  graphics::plot(
    info$f,
    info$P,
    type = 'l',
    main = title_text,
    xlab = 'Frequency',
    ylab = 'P(frequency)',
    col = 'navy',
    lwd = 2
  )
  graphics::abline(v = info$freq, col = 'red', lty = 2, lwd = 2)
}
```









For Shesterkin, the periodograms also fail to show a convincing seasonal cycle. For `cumulative_gsax_ev`, the dominant spike again occurs extremely close to frequency 0, with $f = 0.003509$, so it is better interpreted as long-run trend than seasonality. For `gsax_ev` and `sv_pct_ev`, the strongest peaks correspond to implied periods of about 2.79 and 3 games, but they are only slightly taller than several other peaks, so there is no single isolated spectral spike that clearly dominates the rest of the plot. For `team_xg_pct_ev`, the top peaks are also very close in height to one another, which again argues against a clean seasonal frequency. So for Shesterkin, I also omit harmonic terms and use the same four nonseasonal polynomial trend models.

Shesterkin: Cumulative Goals Saved Above Expected

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.9593	<0.0001	5.6224	3.7848	5.6608
Model II	3	0.9771	<0.0001	5.0492	3.2118	5.1004
Model III	4	0.9773	<0.0001	5.0458	3.2086	5.1098
Model IV	5	0.9812	<0.0001	4.8584	3.0215	4.9353

Model	ME	MPE	MSE	MAE	MAPE
Model I	-8.0143	-13.6854	73.3677	8.0180	13.6916
Model II	0.5612	0.5564	11.2062	2.7151	4.4855
Model III	9.8773	15.8069	152.3764	9.8773	15.8069
Model IV	0.2655	0.2747	6.0653	1.8962	3.1813

Shesterkin: Game-by-Game GSAx

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	-0.0027	0.6283	3.4519	1.6143	3.4903
Model II	3	-0.0062	0.8873	3.4589	1.6215	3.5101
Model III	4	-0.0084	0.8886	3.4645	1.6273	3.5285
Model IV	5	-0.0117	0.9492	3.4712	1.6344	3.5481

Model	ME	MPE	MSE	MAE	MAPE
Model I	0.0612	102.6476	2.3408	1.1972	102.6476
Model II	0.3284	104.5926	2.4582	1.2671	105.7874
Model III	0.0436	98.4712	2.3402	1.1897	100.9516
Model IV	0.3129	108.1113	2.4564	1.2691	108.3716

Shesterkin: Game-by-Game Save Percentage

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	0.0077	0.0744	-2.3627	-4.2003	-2.3243
Model II	3	0.0042	0.2027	-2.3557	-4.1931	-2.3045
Model III	4	0.0047	0.2294	-2.3527	-4.1899	-2.2886
Model IV	5	0.0031	0.3029	-2.3476	-4.1845	-2.2708

Model	ME	MPE	MSE	MAE	MAPE
Model I	-0.0088	-2.4022	0.0081	0.0617	7.8334
Model II	0.0016	-1.2446	0.0081	0.0639	7.9908
Model III	-0.0078	-2.2837	0.0080	0.0616	7.8125
Model IV	-0.0073	-2.2341	0.0080	0.0617	7.8187

Shesterkin: Team xGF%

Model	# Parameters	R_a^2	p_F	AIC	AICc	BIC
Model I	2	-0.0006	0.3671	7.6702	5.8326	7.7086
Model II	3	0.0005	0.3449	7.6725	5.8351	7.7238
Model III	4	-0.0015	0.4623	7.6780	5.8408	7.7420
Model IV	5	0.0069	0.2041	7.6730	5.8362	7.7499

Model	ME	MPE	MSE	MAE	MAPE
Model I	-3.9189	-15.9317	144.9772	9.7920	25.3352
Model II	-4.9320	-18.2224	154.1525	10.0938	26.4045
Model III	1.0844	-4.6721	138.7799	9.4337	22.4344
Model IV	-49.0644	-117.2569	3597.2941	49.9238	118.5210

For Shesterkin's `cumulative_gsax_ev`, the evidence is much cleaner than it was for Blackwood. Model IV is favored by adjusted R^2, AIC, AICc, BIC, MSE, and MAE, so the criteria all point in the same direction. Compared with Blackwood's cumulative series, Shesterkin's smoother upward path is captured very well by the quartic trend model. I therefore select Model IV for Shesterkin's cumulative GSAx. The fitted model is

$$\widehat{CG}_t = -0.069001 + 0.127717t + 0.006883\frac{t^2}{2!} - 0.000132\frac{t^3}{3!} + 0.000001\frac{t^4}{4!}$$

where CG_t is Shesterkin's cumulative even-strength GSAx in goals at career game $t = 1, 2, \ldots, 285$.

For Shesterkin's `gsax_ev`, the deterministic trend signal is again very weak. The overall F-tests are not significant, the information criteria favor the simplest linear specification, and the forecasting differences between Models I and III are tiny. As with Blackwood, the sensible conclusion is that game-by-game GSAx is largely noisy around a stable level rather than strongly trend-driven. So I select Model I for parsimony. The fitted model is

$$\widehat{G}_t = 0.300274 - 0.000471t$$

where G_t is Shesterkin's even-strength game-by-game GSAx in goals at career game t.

For Shesterkin's `sv_pct_ev`, the evidence remains weak under the aligned career-game split. Model I has the largest adjusted R^2 and the smallest AIC, AICc, and BIC, while Model III only barely improves MSE and MAE. Compared with Blackwood, the save-percentage series is a little more stable and shows even less need for higher-order trend terms, so I still select Model I on grounds of parsimony and the very small forecasting differences. The fitted model is

$$\widehat{S}_t = 0.929849 - 0.000095t$$

where S_t is Shesterkin's even-strength save percentage in proportion form at career game t.

For Shesterkin's `team_xg_pct_ev`, the evidence for a deterministic polynomial trend is weak. Model IV has the largest adjusted R^2, but Model I has the smallest AIC, AICc, and BIC, while Model III only slightly improves the holdout MSE and MAE. Since the overall F-tests remain weak and the one-team sample looks comparatively stable, I select Model I as the most sensible summary model for Shesterkin's team xGF%. The fitted model is

$$\widehat{X}_t = 50.100164 - 0.007235t$$

where X_t is Shesterkin's team even-strength xGF% in percentage points at career game t.

In summary, the univariate trend-and-seasonality analysis suggests that neither goalie shows meaningful seasonality in any of the four even-strength series, so harmonic terms are unnecessary here. Blackwood's cumulative GSAx and team xGF% are more visibly shaped by broader regime changes, while Shesterkin's cumulative GSAx is much smoother and more strongly trend-driven over the matched sample. The game-by-game GSAx and save-percentage series for both goalies are much noisier and admit only weak deterministic trends. This part by itself does not prove that GSAx is or is not independent of team context, because I am modeling each series separately rather than directly regressing one on another. However, the comparison does support the original project motivation: a traded goalie and a non-traded goalie can have meaningfully different time-series behavior even under the same even-strength xG framework, so the question of whether GSAx is truly team-independent still deserves further study.